

Лабораторная работа

Номер 6

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Информация

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Развить навыки администрирования ОС Linux. Получить первое практическое знакомство с технологией SELinux. Проверить работу SELinux на практике совместно с веб-сервером Apache.

Проверяем статус SELinux

```
[root@ivanprihodko ~]# getenforce
Enforcing
[root@ivanprihodko ~]# ststratus
bash: ststratus: command not found...
[root@ivanprihodko ~]# sestratus
bash: sestratus: command not found...
[root@ivanprihodko ~]# sestatus
SELinux status:                enabled
SELinuxfs mount:                /sys/fs/selinux
SELinux root directory:         /etc/selinux
Loaded policy name:              targeted
Current mode:                   enforcing
Mode from config file:          enforcing
Policy MLS status:              enabled
Policy deny_unknown status:     allowed
Memory protection checking:     actual (secure)
Max kernel policy version:      33
[root@ivanprihodko ~]# service httpd status
Redirecting to /bin/systemctl status httpd.service
• httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-09-03 05:22:34 MSK; 4min 52s ago
     Docs: man:httpd.service(8)
   Main PID: 5282 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served: 0"
      Tasks: 177 (limit: 101908)
    Memory: 39.9M
       CPU: 269ms
    CGroup: /system.slice/httpd.service
            └─5282 /usr/sbin/httpd -DFOREGROUND
              └─5283 /usr/sbin/httpd -DFOREGROUND
                └─5284 /usr/sbin/httpd -DFOREGROUND
                  └─5287 /usr/sbin/httpd -DFOREGROUND
                    └─5321 /usr/sbin/httpd -DFOREGROUND
```

Находим Apache в списке процессов

```
[root@ivanprihodko ~]# ps auxZ | grep httpd
system_u:system_r:httpd_t:s0  root      5282  0.0  0.0  23328 12620 ?
    Ss   05:22   0:00 /usr/sbin/httpd -DFOREGROUND
system_u:system_r:httpd_t:s0  apache  5283  0.0  0.0  23152  5116 ?
    S    05:22   0:00 /usr/sbin/httpd -DFOREGROUND
system_u:system_r:httpd_t:s0  apache  5284  0.0  0.1 2424300 17888 ?
    Sl   05:22   0:00 /usr/sbin/httpd -DFOREGROUND
system_u:system_r:httpd_t:s0  apache  5287  0.0  0.0 2227628 15600 ?
    Sl   05:22   0:00 /usr/sbin/httpd -DFOREGROUND
system_u:system_r:httpd_t:s0  apache  5321  0.0  0.0 2162092 15468 ?
    Sl   05:22   0:00 /usr/sbin/httpd -DFOREGROUND
unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 root  5670  0.0  0.0 22181
2 2432 pts/0 S+ 05:30   0:00 grep --color=auto httpd
```

Рис. 2: Поиск Apache в списке процессов

Проверяем текущее состояние переключателей SELinux

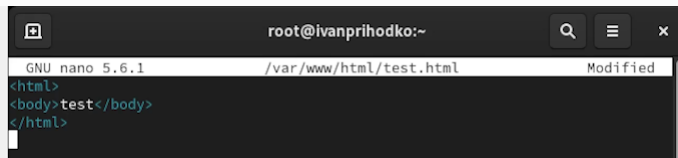
```
[root@ivanprihodko ~]# sestatus -b | grep httpd
httpd_anon_write                off
httpd_builtin_scripting         on
httpd_can_check_spam            off
httpd_can_connect_ftp           off
httpd_can_connect_ldap          off
httpd_can_connect_mythtv        off
httpd_can_connect_zabbix        off
httpd_can_manage_courier_spool   off
httpd_can_network_connect       off
httpd_can_network_connect_cobbler off
httpd_can_network_connect_db    off
httpd_can_network_memcache      off
httpd_can_network_relay         off
httpd_can_sendmail              off
httpd_dbus_avahi                off
httpd_dbus_sssd                 off
httpd_dontaudit_search_dirs     off
httpd_enable_cgi                on
httpd_enable_ftp_server         off
httpd_enable_homedirs           off
httpd_execmem                   off
httpd_graceful_shutdown         off
httpd_manage_ipa                off
httpd_mod_auth_ntlm_winbind     off
httpd_mod_auth_pam              off
httpd_read_user_content         off
httpd_run_ipa                   off
httpd_run_preupgrade            off
```

Проверяем информацию о SELinux

```
[root@ivanprihodko ~]# seinfo
Statistics for policy file: /sys/fs/selinux/policy
Policy Version:          33 (MLS enabled)
Target Policy:           selinux
Handle unknown classes:  allow
Classes:                 135   Permissions:         457
Sensitivities:           1     Categories:          1024
Types:                   5186  Attributes:           259
Users:                   8     Roles:                15
```

Рис. 4: Проверка информации о SELinux

Создаем тестовый html файл



The screenshot shows a terminal window with a dark background. The title bar at the top reads 'root@ivanprihodko:~'. Below the title bar, the terminal shows the GNU nano 5.6.1 editor interface. The file being edited is '/var/www/html/test.html', and it is marked as 'Modified'. The content of the file is as follows:

```
<html>
<body>test</body>
</html>
```

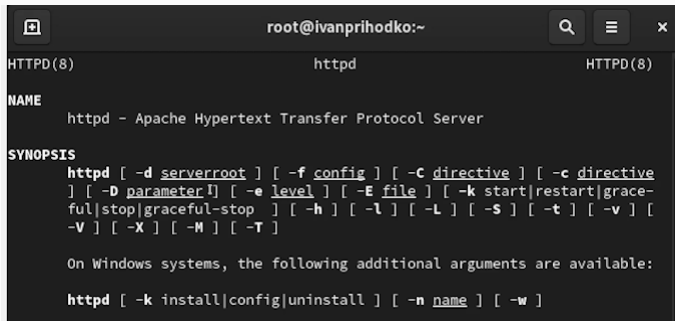
Рис. 5: Создание тестовых html файлов

Изучение справки httpd_selinux

```
[root@ivanprihodko ~]# ls -lZ /var/www
total 0
drwxr-xr-x. 2 root root system_u:object_r:httpd_sys_script_exec_t:s0 6 Aug 26
02:00 cgi-bin
drwxr-xr-x. 2 root root system_u:object_r:httpd_sys_content_t:s0 6 Aug 26
02:00 html
[root@ivanprihodko ~]# ls -lZ /var/www/html
total 0
[root@ivanprihodko ~]# touch /var/www/html/test.html
[root@ivanprihodko ~]# nano /var/www/html/test.html
[root@ivanprihodko ~]# ls -lZ /var/www/html
total 4
-rw-r--r--. 1 root root unconfined_u:object_r:httpd_sys_content_t:s0 33 Sep 3
05:34 test.html
[root@ivanprihodko ~]# man httpd_selinux
No manual entry for httpd_selinux
[root@ivanprihodko ~]# man httpd
```

Рис. 6: Изучение справки httpd_selinux

Изучение справки httpd_selinux



```
root@ivanprihodko:~  
HTTPD(8)                                httpd                                HTTPD(8)  
  
NAME  
    httpd - Apache Hypertext Transfer Protocol Server  
  
SYNOPSIS  
    httpd [ -d serverroot ] [ -f config ] [ -C directive ] [ -c directive  
    ] [ -D parameter ] [ -e level ] [ -E file ] [ -k start|restart|grace-  
    ful|stop|graceful-stop ] [ -h ] [ -l ] [ -L ] [ -S ] [ -t ] [ -v ] [ -V ] [ -X ] [ -M ] [ -T ]  
  
    On Windows systems, the following additional arguments are available:  
  
    httpd [ -k install|config|uninstall ] [ -n name ] [ -w ]
```

Рис. 7: Изучение справки httpd_selinux

Обращаемся к тестовому html файлу

```
[root@ivanprihodko ~]# ls -Z /var/www/html/test.html
unconfined_u:object_r:httpd_sys_content_t:s0 /var/www/html/test.html
[root@ivanprihodko ~]# ls -Z /var/www/html/test.html
unconfined_u:object_r:httpd_sys_content_t:s0 /var/www/html/test.html
[root@ivanprihodko ~]# man /var/www/html/test.html
[root@ivanprihodko ~]# chcon -t samba_share_t /var/www/html/test.html
[root@ivanprihodko ~]# ls -Z /var/www/html/test.html
unconfined_u:object_r:samba_share_t:s0 /var/www/html/test.html
[root@ivanprihodko ~]# ls -l /var/www/html/test.html
-rw-r--r--. 1 root root 33 Sep  3 05:34 /var/www/html/test.html
[root@ivanprihodko ~]#
```

Рис. 8: Обращение к тестовому html файлу

Проверка логов

```
.html to public_content_t or public_content_rw_t.#012Do#012# semanage fcontext
-a -t public_content_t '/var/www/html/test.html'#012# restorecon -v '/var/www
/html/test.html'#012#012***** Plugin catchall (1.41 confidence) suggests **
*****#012#012If you believe that httpd should be allowed ge
tattr access on the test.html file by default.#012Then you should report this
as a bug.#012You can generate a local policy module to allow this access.#012D
o#012allow this access for now by executing:#012# ausearch -c 'httpd' --raw |
audit2allow -M my-httpd#012# semodule -X 300 -i my-httpd.pp#012
Sep  3 05:38:01 ivanprihodko systemd[1]: dbus-:1.1-org.fedoraproject.Setrouble
shootPrivileged@1.service: Deactivated successfully.
Sep  3 05:38:01 ivanprihodko systemd[1]: dbus-:1.1-org.fedoraproject.Setrouble
shootPrivileged@1.service: Consumed 1.324s CPU time.
Sep  3 05:38:01 ivanprihodko systemd[1]: setroubleshootd.service: Deactivated
successfully.
Sep  3 05:38:01 ivanprihodko systemd[1]: setroubleshootd.service: Consumed 1.1
56s CPU time.
[root@ivanprihodko ~]#
```

Рис. 9: Проверка логов

Тестовый файл работает успешно

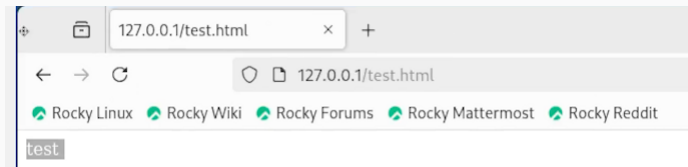


Рис. 10: Работа тестового файла

Меняем конфиг файл

```
[root@ivanprikhodko ~]# nano /etc/httpd/httpd.conf
[root@ivanprikhodko ~]# tail -nl /var/log/messages
tail: invalid number of lines: 'l'
[root@ivanprikhodko ~]# tail -n /var/log/messages
tail: invalid number of lines: '/var/log/messages'
[root@ivanprikhodko ~]# cat /var/log/http/error_log
cat: /var/log/http/error_log: No such file or directory
[root@ivanprikhodko ~]# cat /var/log/audit/audit.log
```

Рис. 11: Изменение конфиг файла

Проверка списка портов

```
[root@ivanprihodko ~]# semanage port -a -t http_port_t -p tcp 81
usage: semanage [-h]
               {import,export,login,user,port,ibpkey,ibendport,interface,module,node,fcontext,boolean,permissive,dontaudit}
               ...
semanage: error: unrecognized arguments: -p 81
[root@ivanprihodko ~]# semanage port -l | grep http_port_t
http_port_t          tcp          80, 81, 443, 488, 8008, 8009, 8443, 9000
pegasus_http_port_t  tcp          5988
[root@ivanprihodko ~]# chcon +t samba_share_t /var/www/html/test.html
chcon: invalid context: '+t': Invalid argument
[root@ivanprihodko ~]# chcon t samba_share_t /var/www/html/test.html
chcon: invalid context: 't': Invalid argument
[root@ivanprihodko ~]# chcon samba_share_t /var/www/html/test.html
chcon: invalid context: 'samba_share_t': Invalid argument
[root@ivanprihodko ~]# chcon -t httpd_sys_content_t /var/www/html/test.html
[root@ivanprihodko ~]# nano /etc/httpd/httpd.conf
[root@ivanprihodko ~]# semanage port -d -t http_port_t -p tcp 81
ValueError: Port tcp/81 is defined in policy, cannot be deleted
[root@ivanprihodko ~]# sudo semanage port -d -t http_port_t -p tcp 81
ValueError: Port tcp/81 is defined in policy, cannot be deleted
[root@ivanprihodko ~]# rm /var/www/html/test.html
rm: remove regular file '/var/www/html/test.html'? yes
[root@ivanprihodko ~]#
```

Рис. 12: Проверка списка портов

После всего этого мы не можем зайти на текстовый сайт

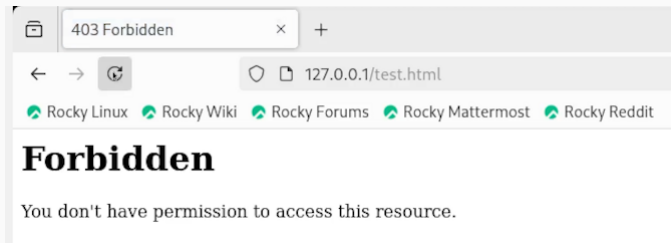


Рис. 13: Неудачный заход на тестовый сайт

В ходе данной лабораторной работы были получены знания для работы с html файлами